

Danmarks
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Report Applied Hydrodynamics: Study of a Ship on HydroD

AUTHORS

Maxime Becuwe - s243496

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Abstract

This project investigates the hydrodynamic response and sectional loads of a 200 m ship using HydroD. Zero-speed analyses with Wadam and forward-speed analyses with Wasim at 6, 8 and 10 m/s examine vessel motions, roll damping, wave-induced loads and added resistance. Response amplitude operators are combined with a Pierson–Moskowitz wave spectrum to estimate time histories and short-term extreme loads, complemented by a long-term assessment at zero speed. The reported simulations show reduced roll response with additional damping and a speed-dependent pitch response. Sectional-load and added-resistance estimates provide inputs for subsequent structural and propulsion studies. The assessment remains subject to numerical convergence, consistent phase and heading conventions, and verification of the original extreme-value calculations. The deck-wetness assessment is limited to one simulated realization and does not establish an operational safety limit.

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1 Introduction

1.1 Objectives

This report aims to analyze the hydrodynamic behavior and structural loads of a 200-meter ship using HydroD. The study is divided into two parts: Wadam (steady ship analysis) and Wasim (analysis with forward speed at 6, 8, and 10 m/s).

1.2 Description of the ship and the environment

1.2.1 Main dimension of the ship

The studied ship is 200 m long and 32 m wide. The designed draft is 11.25 m, a value used to determine its displacement. Thus, this ship has a displacement of $M = 60$ kilotons.

1.2.2 Mass distribution

An appropriate mass distribution is essential for analysing the ship, particularly its sectional loads. Although discrete point masses provide an initial approximation in HydroD, a distributed mass profile provides a more representative basis for this study.

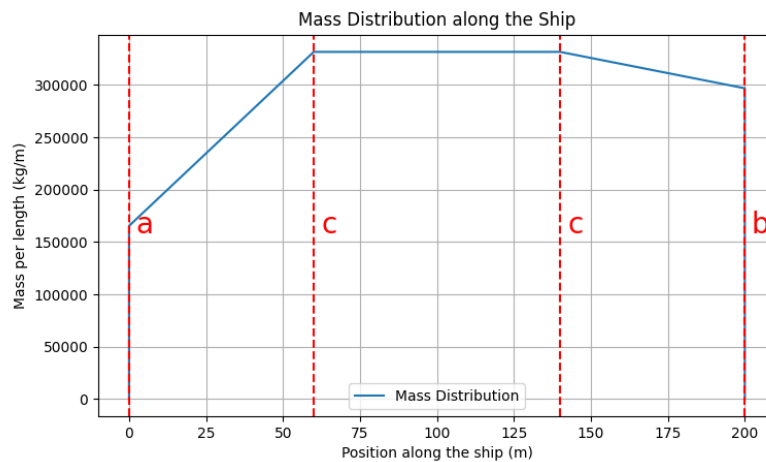


Figure 1: mass distribution of the ship

How to define the mass distribution?

The mass profile shown in Figure 1 is defined by three unknowns: a , b , and c . One constraint is known: the integral of the mass profile must equal the displacement of the ship.

$$0.5 * 0.3L * [(a + c) + (b + c)] + 0.4Lc = M \quad (1.1)$$

Let s run from 0 to L along the plotted profile, and let $x = s - L/2$ denote the coordinate measured from midship. For the piecewise-linear distribution shown, mass and first-moment

conservation give

$$M = L(0.15a + 0.15b + 0.70c), \quad (1.2)$$

$$Ms_G = L^2(0.015a + 0.135b + 0.35c), \quad s_G = x_G + L/2. \quad (1.3)$$

With $\gamma = s_G/L$ and $\alpha = cL/M$, the solution is

$$a = \frac{9 - 10\gamma - 2.8\alpha}{1.2} \frac{M}{L}, \quad b = \frac{10\gamma - 2.8\alpha - 1}{1.2} \frac{M}{L}, \quad c = \alpha \frac{M}{L}. \quad (1.4)$$

The original report specifies $x_G = 2.23\text{m}$ and $\alpha = 1.1$. Interpreting this x_G relative to midship gives $\gamma = 0.51115$. The plotted profile is retained as originally exported; its end-point values should be checked against the actual model because the original algebra mixed coordinate origins and omitted a factor of L in the first moment.

This piecewise-linear mass distribution is represented by discrete masses distributed along the ship in two longitudinal lines. Each mass will be defined by its x -position, calculated based on the previously detailed mass distribution. To represent it in HydroD, the mass must then be defined as two lines in the same XY plane. All points will be located in this plane, with a z -coordinate of -4.25m (value provided by a user of this ship). The Y -coordinates will be set to the R_x value given by the same owner, to represent the specified transverse radius of gyration. A radius of gyration is a length, not a moment of inertia; the target roll inertia is $I_{xx} = MR_x^2$ about the centre of gravity, and the complete mass distribution must reproduce it.

1.2.3 Seastate

The seastate will be defined by a Pierson-Moskowitz (PM) spectrum. The Spectral Energy Density is given by the following formula:

$$S(\omega) = \frac{5}{16} H_s^2 \omega_p^4 \omega^{-5} e^{-1.25(\frac{\omega_p}{\omega})^4} \quad (1.5)$$

Where H_s is the significant wave height, and T_p is the peak period.

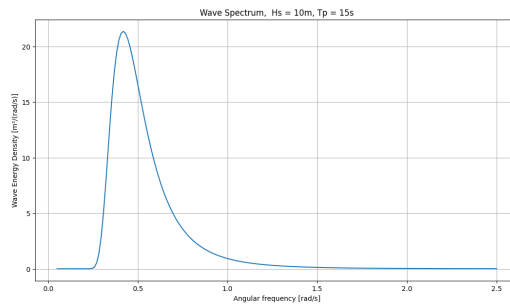


Figure 2: Example of a PM wave spectrum, with $H_s = 10\text{m}$ and $T_p = 15\text{s}$

2 theory

This section presents the theories used in Wadam, Wasim and the post-processing procedures.

2.1 Introduction of Wadam

The Wadam algorithm works on solving the potential flow problem around a floating or fixed structure. The algorithm uses the Boundary Element Method (BEM) to discretize the problem.

Potential theory is applied in Wadam to calculate the first-order radiation and diffraction effects on large-volume structures. This approach uses a 3D panel method to evaluate velocity potentials and hydrodynamic coefficients.

First-order potential theory linearizes the boundary conditions about the mean geometry. Second-order theory retains terms of second order in wave amplitude; it is a perturbation approximation and should not be described as imposing the fully nonlinear free-surface condition without approximation.

The radiation and diffraction velocity potentials on the wet surface of the body are determined by solving an integration equation derived from Green's theorem. The source strengths are calculated using the source distribution method, relying on the same potentials to represent the fluid motion

Numerically, the body surface is approximated using planar quadrilateral panels, and the integral equation is discretized into a system of algebraic equations. This discretization enables the computation of velocity potentials and hydrodynamic coefficients needed for analyzing the interaction between the structure and the waves.

- Geometry discretisation: the structure is discretized into a mesh of panels, essential to use the BEM method.
- Potential flow assumptions: the fluid is assumed to be incompressible, inviscid and irrotational. These assumptions simplify the governing equations.
- Boundary conditions enforce the free-surface conditions and prevent flow through the hull and seabed.
- The velocity potential satisfies Laplace's equation in the fluid domain.

2.2 Wasim

Wasim is a program for computing global responses of ships with forward speed within the displacement-vessel modelling assumptions. The simulations are carried out in the time domain, and frequency-domain results can be obtained using a Fourier transform.

Wasim solves the fully 3-dimensional radiation/diffraction problem by a Rankine panel method. For these methods panels are required both on hull and on the free surface (for

Wadam it was only on the hull).

The solution method is the Rankine Panel Method. The difference between a Rankine Panel Method and a Green's function based panel method (Wadam) is that the elementary solution in the Rankine Method does not satisfy the free surface boundary condition.

Wasim uses the Time-Domain Approach because it is more flexible for including non-linear effects such as nonlinear drag loads.

2.3 The potential flow theory

Both Wadam and Wasim use the potential flow theory. In this theory, the velocity of the fluid is derived from a velocity potential.

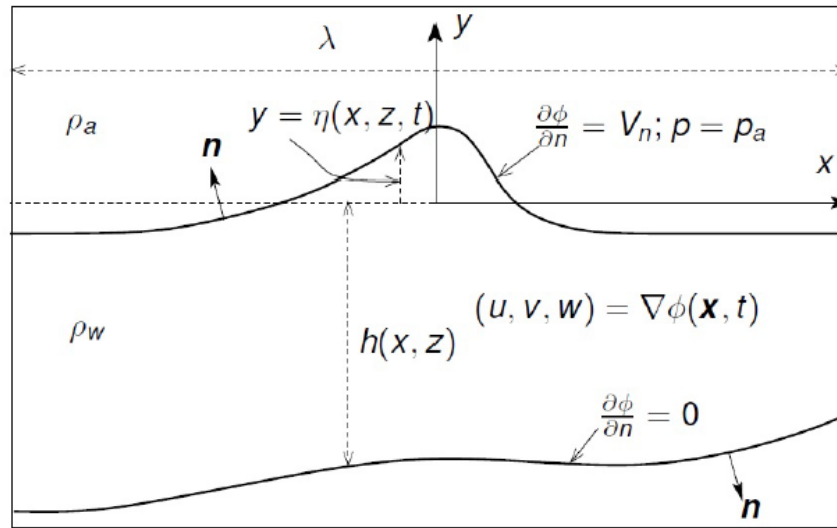


Figure 3: Sketch of a wave and the boundary conditions on free surface and sea floor

Let z point upward, with the undisturbed free surface at $z = 0$. For constant density, incompressibility gives

$$\nabla \cdot \mathbf{v} = 0, \quad \mathbf{v} = \nabla \Phi, \quad (2.1)$$

where irrotational flow permits the velocity potential Φ . Thus $\nabla^2 \Phi = 0$ in the fluid. For inviscid, irrotational flow under gravity, Bernoulli's equation is

$$\frac{\partial \Phi}{\partial t} + \frac{1}{2} |\nabla \Phi|^2 + \frac{p}{\rho} + gz = C(t). \quad (2.2)$$

The potential can absorb the time-dependent constant. With atmospheric pressure as the

pressure datum and neglecting surface tension, the exact free-surface conditions are

$$\eta_t + \Phi_x \eta_x + \Phi_y \eta_y = \Phi_z, \quad z = \eta(x, y, t), \quad (2.3)$$

$$\Phi_t + \frac{1}{2} |\nabla \Phi|^2 + g\eta = 0, \quad z = \eta(x, y, t). \quad (2.4)$$

No penetration requires $\partial \Phi / \partial n = V_n$ on the moving hull and $\partial \Phi / \partial n = 0$ on a fixed seabed.

For small-amplitude waves at zero forward speed, the first-order problem is

$$\nabla^2 \Phi = 0, \quad -h < z < 0, \quad (2.5)$$

$$\eta_t = \Phi_z, \quad z = 0, \quad (2.6)$$

$$\Phi_t + g\eta = 0, \quad z = 0, \quad (2.7)$$

$$\Phi_z = 0, \quad z = -h. \quad (2.8)$$

These equations use a horizontal seabed at depth h . Forward-speed analysis requires the corresponding moving-frame formulation; it is not obtained by using the zero-speed equations unchanged.

With $\Phi(\mathbf{x}, t) = \text{Re}\{\phi(\mathbf{x})e^{i\omega t}\}$, the combined free-surface condition is

$$\phi_z - \frac{\omega^2}{g} \phi = 0, \quad z = 0. \quad (2.9)$$

Define radiation potentials ϕ_j per unit generalized body velocity. If ξ_j are complex motion amplitudes, then

$$\phi = \phi_0 + \phi_D + i\omega \sum_{j=1}^6 \xi_j \phi_j. \quad (2.10)$$

Here ϕ_0 is the incident potential and ϕ_D the scattered potential. On the mean wetted hull,

$$\frac{\partial \phi_j}{\partial n} = n_j, \quad \frac{\partial(\phi_0 + \phi_D)}{\partial n} = 0, \quad (2.11)$$

where n_j is the generalized normal component for motion j . For incident wave amplitude A and propagation angle β measured from the positive x axis,

$$\phi_0 = \frac{igA}{\omega} \frac{\cosh k(z+h)}{\cosh kh} e^{-ik(x \cos \beta + y \sin \beta)}, \quad \omega^2 = gk \tanh(kh). \quad (2.12)$$

In deep water, this reduces to $\omega^2 = gk$. These governing relations follow standard linear water-wave theory [1].

2.4 Boundary element method

Green's second identity converts Laplace's equation into a boundary integral relation. With $G(\boldsymbol{\xi}, \mathbf{x}) = 1/(4\pi|\boldsymbol{\xi} - \mathbf{x}|)$ and the normal directed out of a bounded fluid domain Ω , a

representative form is

$$c(\mathbf{x})\phi(\mathbf{x}) = \int_{\partial\Omega} \left[G \frac{\partial\phi}{\partial n} - \phi \frac{\partial G}{\partial n} \right] dS. \quad (2.13)$$

At a smooth boundary, $c = 1/2$. This generic identity includes the complete domain boundary; an exterior wave problem also requires a radiation condition. A free-surface Green function incorporates additional boundary conditions and changes the surfaces that must be discretized. Wadam uses this approach, whereas a Rankine-source formulation discretizes the free surface explicitly. Panel discretization converts the applicable integral equations into linear algebraic systems. Signs and numerical prefactors depend on the chosen normal and Green-function normalization, so formulations with different conventions must not be mixed.

2.5 Second order non linear wave loads

When it is exposed to waves, the ship experiences wave loads at the same frequencies as the waves. In reality, there are also non linear effects causing non linear wave loads and responses: mean wave loads, low frequency wave loads and sum-frequency wave loads:

- **mean wave loads:** there are nonlinear steady forces resulting from time-averaged wave pressures over the structure's wetted surface.
- **low-frequency wave loads:** Second-order forces caused by the interaction of wave components with close frequencies, leading to difference-frequency loads. These forces excite slow-drift motions, significant in horizontal direction (surge sway yaw).
- **Sum-frequency wave loads:** high frequency forces are caused by interactions between wave components of different frequencies, leading to sum-frequency response.

2.6 Mean drift force and Added wave resistance

Mean drift forces are second-order forces resulting from the integration of pressure over the wetted surface of the structure. These forces are due to the asymmetry of wave-induced pressures. Second order wave solution around floating structures are due to nonlinear interaction between linear (incident, diffraction and radiation) wave fields. Both second-order potential and the quadratic pressure on the hull and at the waterline can contribute to the second order wave loads.

In Wadam, the mean drift force is obtained by far-field integration. This approach applies momentum conservation in the control volume enclosing the structure, focusing on waves radiated or diffracted.

The added wave resistance is mean-drift along the longitudinal direction (surge direction) of the ship. When the ship travel against wave (heading = 180), the mean-drift is in the opposite direction as the forward speed, representing a resistance that the ship has to overcome through extra power compared to calm water.

2.7 Short term and long term analysis

Short- and long-term analyses estimate the most probable maximum responses, particularly the sectional loads.

2.7.1 short term analysis

Short-term analysis relies on the assumption that the sea state is stationary over a relatively short period, typically on the order of a few hours. During this time, the environmental conditions (e.g., significant wave height, period, etc.) are considered constant.

The objective of the short term analysis is to predict the probable load under a specific set of environmental conditions (seastate fixed by a PM spectrum, see 1.2.3) during a short time (3 hours) for a given direction.

During this study, $\omega_p = \frac{2\pi}{T_p}$ is the peak frequency, and H_s is the Significant Wave Height: $H_s = 4\sqrt{m_0}$ and $m_0 = \int_0^\infty S_{wave}(\omega)d\omega$

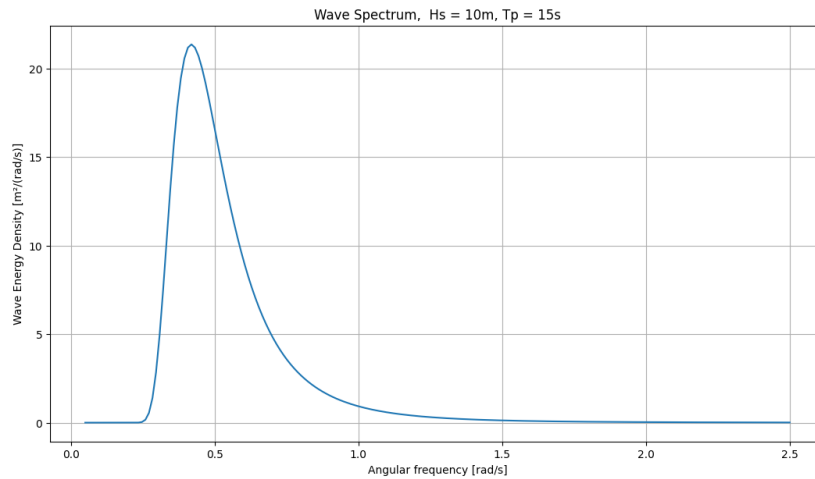


Figure 4: Wave spectrum used for the short term analysis: $H_s = 10m$, $T_p = 15s$

Response: The response spectrum of Response is equal to:

$$S_R = |H_R|^2 S_{wave} \quad (2.14)$$

Probability density function: For a stationary, zero-mean, narrow-band Gaussian response, positive peak amplitudes are approximated by a Rayleigh distribution. This is a distribution of individual peaks, not of the maximum over an entire record:

$$f_p(r_p) = \frac{r_p}{\sigma_R^2} e^{-\frac{r_p^2}{2\sigma_R^2}} \quad (2.15)$$

$r_p \geq 0$ is a peak amplitude. For a large number of approximately independent peaks, the commonly used most-probable-maximum approximation is [2]

$$R_M(H_s, T_p) \approx \sigma_R \sqrt{2 \ln N} \quad (2.16)$$

$$\sigma_R = \sqrt{m_0} \quad \text{where } m_0 = \int_0^\infty S_R d\omega \quad (2.17)$$

Here $N \approx T_d/T_{z,R}$, with $T_d = 10800$ s for a three-hour assessment. For a response spectrum expressed per unit angular frequency,

$$m_{n,R} = \int_0^\infty \omega^n S_R(\omega) d\omega, \quad T_{z,R} = 2\pi \sqrt{\frac{m_{0,R}}{m_{2,R}}}, \quad \nu_{0,R} = \frac{1}{T_{z,R}}. \quad (2.18)$$

The response zero-upcrossing period is not generally the wave-spectrum zero-upcrossing period. The factor 2π follows from using angular frequency rather than frequency in hertz.

2.7.2 Long Term Analysis

Long-term analysis combines sea-state exposure with conditional response statistics. Let p_{ij} be the fraction of operational time in sea-state bin (i, j) , with $\sum p_{ij} = 1$, and let $T_{z,R,ij}$ and $\sigma_{R,ij}$ describe the response in that bin. For duration T_L , the approximate number of response peaks is $N_{ij} = T_L p_{ij} / T_{z,R,ij}$. Under the independent-peak Rayleigh approximation, the lifetime-maximum distribution is

$$F_{\max}(r) = \prod_{i,j} \left[1 - \exp\left(-\frac{r^2}{2\sigma_{R,ij}^2}\right) \right]^{N_{ij}}, \quad r \geq 0. \quad (2.19)$$

Only occupied bins with nonzero response variance are included. The most probable lifetime maximum is the mode of $dF_{\max}/dr$; an interior candidate satisfies $d^2 F_{\max}/dr^2 = 0$ and must correspond to a density maximum. A time-weighted mixture of individual-peak distributions alone does not specify a 20-year maximum. Heading exposure and the conversion between the scatter diagram's wave period and the spectrum's peak period must also be specified. The original long-term output is retained below, but its exposure weighting and implementation require verification.

Significant wave height [m]	Peak period [s]																			
	2.5	3.5	4.5	5.5	6.5	7.5	8.5	9.5	10.5	11.5	12.5	13.5	14.5	15.5	16.5	17.5	18.5	19.5	20.5	
0.5	42	429	1172	1662	1634	1291	888	558	331	189	105	57	31	17						
1.5	2	109	964	3034	5260	6292	5902	4689	3317	2160	1325	778	443	246	135	73	39	21	11	
2.5		2	76	578	1837	3373	4297	4248	3504	2532	1659	1009	581	320	171	89	46	23	11	
3.5			2	47	320	1006	1881	2442	2436	2000	1419	900	523	285	147	73	35	16	7	
4.5				2	33	206	635	1181	1519	1485	1179	797	475	257	128	60	27	12	5	
5.5					1	22	133	395	715	888	829	619	388	212	103	46	19	7	3	
6.5						1	16	85	242	414	485	420	287	163	79	34	13	5	2	
7.5							1	11	55	144	228	244	191	117	58	25	9	3	1	
8.5								1	8	35	83	117	112	77	41	18	6	2	1	
9.5									1	6	22	45	55	45	27	12	4	1		
10.5										1	4	13	22	23	16	8	3	1		
11.5											1	3	7	10	9	5	2	1		
12.5													2	4	4	3	1			
13.5														1	2	1	1			
14.5																1				

Figure 5: Example of a Scatter Diagram

3 Steady ship - Wadam

3.1 Parameters in Wadam

3.1.1 simulation parameters

The parameters in Wadam are defined as follows:

- Frequency Set: from 0.2 to 2.5 rad/s, with a step of 0.1 rad/s.
- Direction Set: from 0 to 360 deg, with a step of 15 deg.
- location: infinity borders, depth = 300m.
- for air: $g = 9.80665 m/s^2$, $\rho_{air} = 1.226 kg/m^3$, $\nu_{air} = 1.462e - 005 m^2/s$
- for water: $\rho_{water} = 1025 kg/m^3$, $\nu_{water} = 1.19e - 006 m^2/s$
- drift forces computed by far field integration.
- numerical integration: one node Gauss

3.1.2 mesh

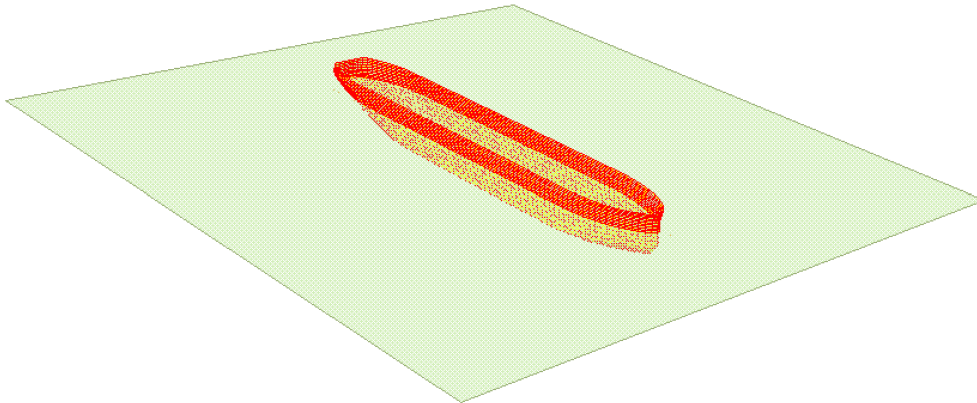


Figure 6: Mesh of the ship and the loading condition

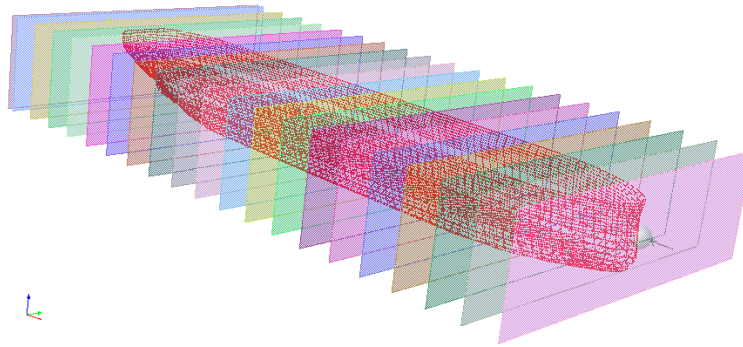
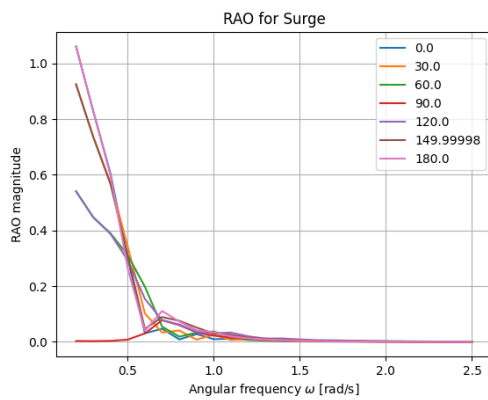


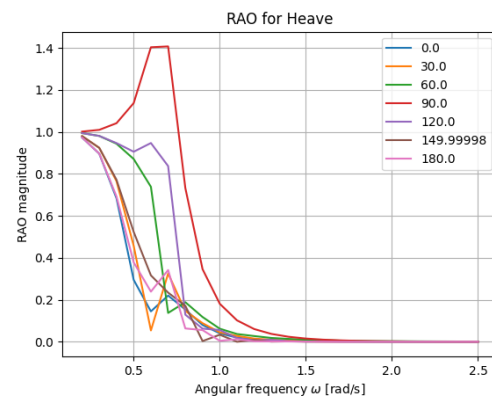
Figure 7: Mesh with the different cross sections.

3.2 ship response

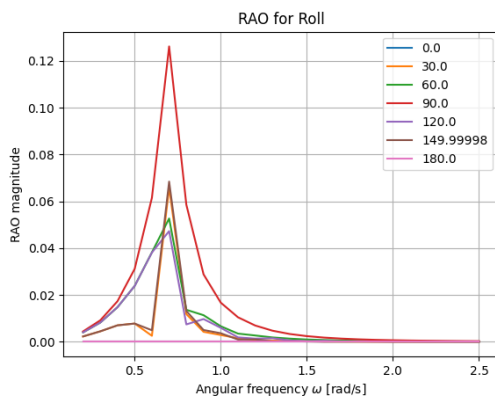
3.2.1 RAOs



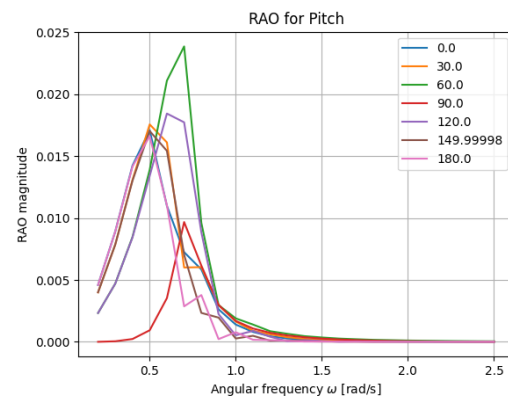
(a) Surge motion



(b) Heave motion



(c) Roll motion



(d) Pitch motion

Figure 8: RAO of the ship for different motion.

The four plots show different response maxima: **Heave and roll:** the plotted beam-sea responses reach pronounced peaks near $\omega = 0.7 \text{ rad/s}$, corresponding to $T = 2\pi/\omega \approx 8.98 \text{ s}$. The deep-water relation $\omega^2 = gk$ gives $\lambda = 2\pi g/\omega^2 \approx 126 \text{ m}$, approximately 0.63 times the ship length. Peak amplitudes depend on excitation, restoring forces, inertia and damping; wavelength alone does not establish their cause.

Surge: the plotted surge response is largest at the low-frequency end of the evaluated range. A displacement RAO measures response per unit wave amplitude; it is not a direct measure of absorbed wave energy.

3.2.2 temporal response of the ship

The ship's RAOs for the six motions (surge, sway, heave, roll, pitch and yaw) can be combined with a specified wave spectrum to calculate time-domain responses.

The time-domain solution takes the following form:

$$\eta(t) = \sum_{j=1}^N A_{wave}^j \cos(\omega_j t + \phi_{wave}^j) \quad (3.1)$$

$$R_{ship}(t) = \sum_{j=1}^N A_{ship}^j \cos(\omega_j t + \phi_{ship}^j) \quad (3.2)$$

The frequency domain is discretized into intervals of $\Delta\omega$, and the corresponding amplitude and phase are calculated for each ω_j .

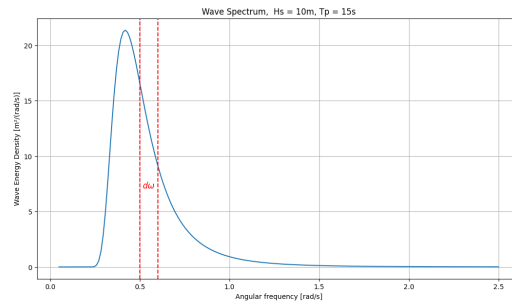


Figure 9: discretization of the wave spectrum

The Amplitude of the waves for each frequencies is equal to $A_{wave}^j = \sqrt{2S(\omega_j)d\omega}$ and the phase is defined randomly between 0 and 2π .

The amplitudes and the phases of the ship can be obtained as follow: $A_{ship}^j = |RAO(\omega_j)| \cdot A_{wave}^j$ and $\phi_{ship}^j = \phi_{wave}^j + \arg(RAO(\omega_j))$, for a complex RAO defined with the same $e^{i\omega t}$ convention

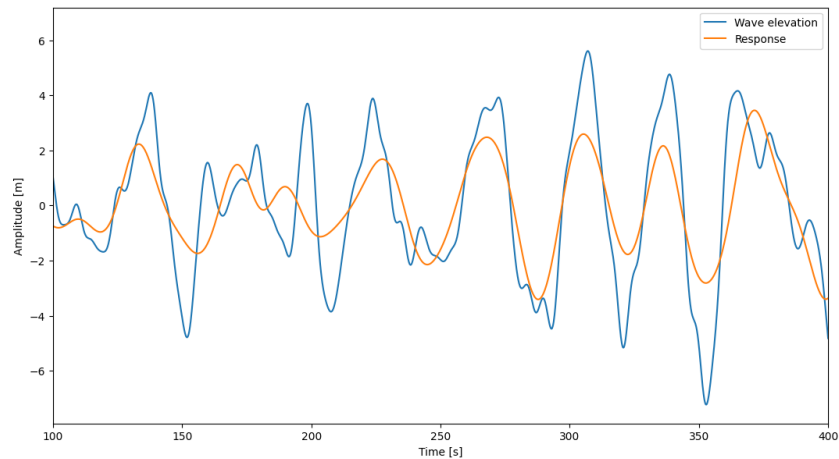


Figure 10: temporal response of the steady ship

3.3 Effect of the non linear roll damping term on the ship

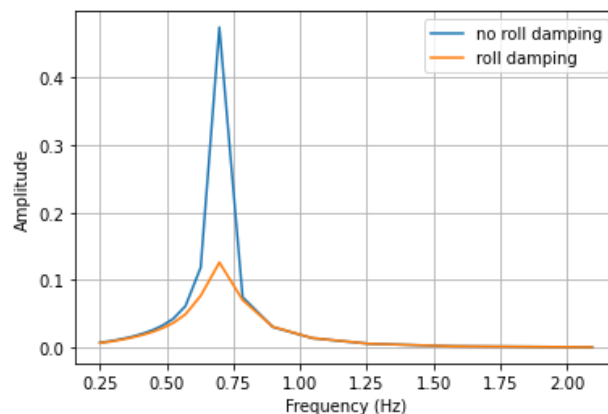


Figure 11: Amplitude of roll motion, with and without a roll damping term

The non linear roll damping term is an additional term included in the roll motion equation to account for viscous effects. The preceding figure shows a substantially smaller resonance amplitude when the roll damping term accounts for viscous effects.

When nonlinear roll damping is represented by an equivalent linear coefficient, that coefficient can depend on the assumed response amplitude and sea state. The resulting effective RAO can therefore vary between sea states, unlike an RAO from a fixed linear model. This interpretation requires consistent damping settings when comparing runs.

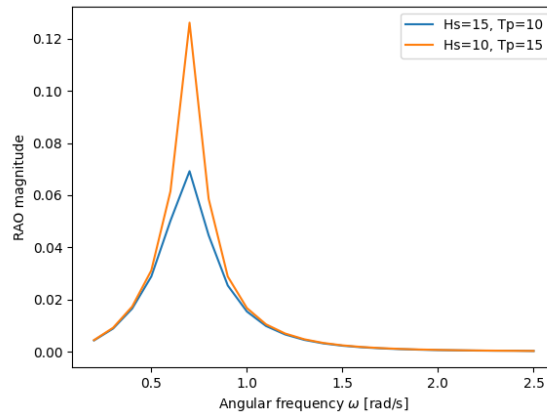


Figure 12: comparison of the RAO for roll motion for 2 different SeaState

The nonlinear roll damping term takes care of the viscous effect. It depends strongly on the angular velocity and, therefore, on the amplitude of the roll motion. This amplitude, in turn, is influenced by the wave spectrum of the seastate:

- For a larger H_s : The significant wave height increases the roll excitation, leading to greater motions and, consequently, an increase in dissipative forces.
- For a longer T_p : The effect depends on whether the spectral peak moves toward or away from roll resonance; increasing the peak period does not necessarily increase roll response.

More generally, it can be observed that the amplitudes remain very small when taking into account the non-linear roll damping term (the maximum amplitude being less than 0.12).

3.4 effect of a bilge keel

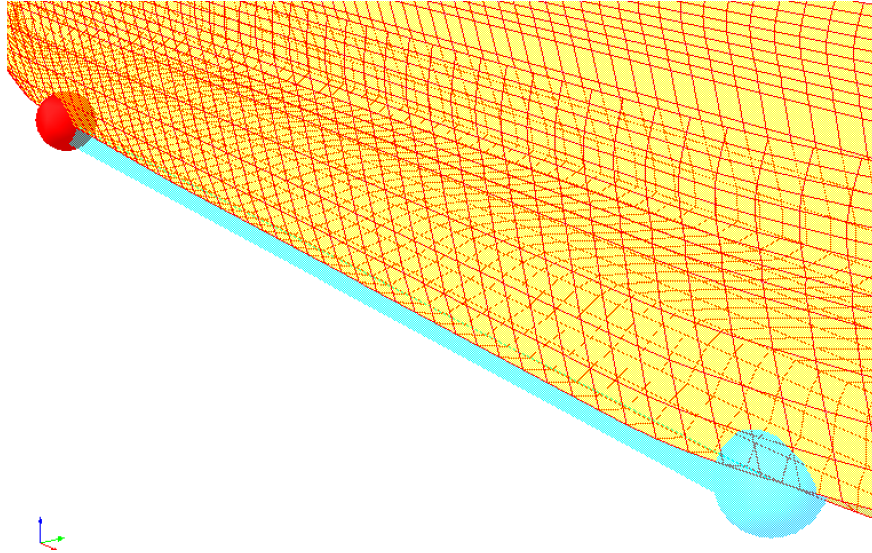


Figure 13: Bilge on the ship

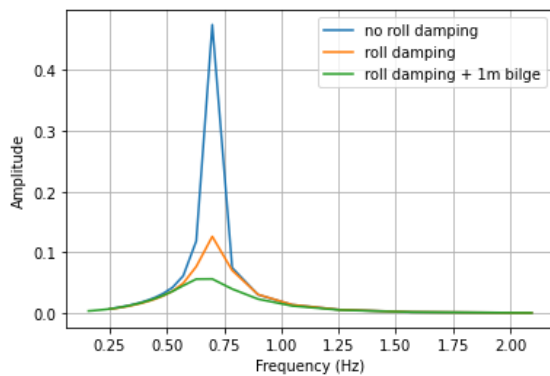


Figure 14: effect of a bilge keel on roll motion

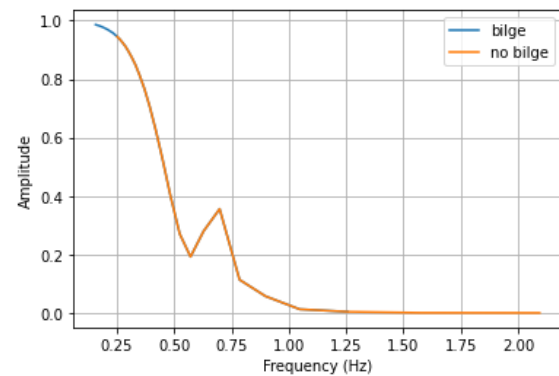


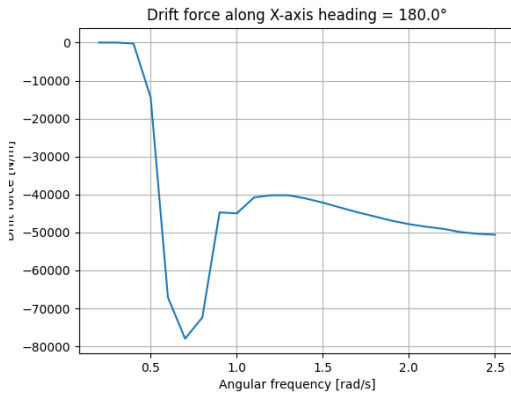
Figure 15: effect of a bilge keel on heave motion

As shown in the 2 figures above, the bilge keel reduces the roll motion of the ship. This makes the ship safer and more comfortable for the crew in sea states with frequencies close to resonance. In contrast, for the heave motion, it can be observed that the addition of a bilge keel does not have an impact on the RAO of the heave motion.

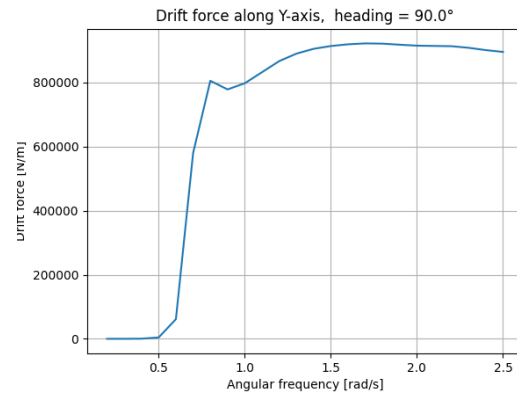
From a structural perspective, the reduced roll response suggests lower roll-induced accelerations and associated inertial loading on equipment and cargo for the conditions considered. This benefit should not be interpreted as a demonstrated reduction in vertical hull-girder bending, since the heave response is essentially unchanged and the sectional loads have not been compared here. Bilge keels also transfer hydrodynamic forces into the

shell through their attachments, making local reinforcement and fatigue-resistant connection details relevant. Their structural benefit therefore involves both reduced vessel motion and adequate resistance at the attachment locations.

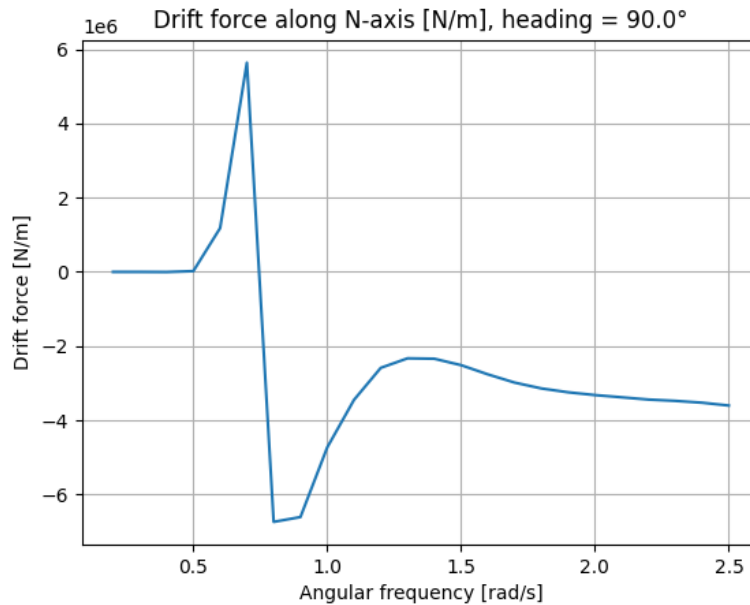
3.5 drift forces



(a) Drift force along X-axis for heading = 180°



(b) Drift force along Y-axis for heading = 90°



(c) Drift force rotation around Z for heading = 90°

Figure 16: Drift forces for a steady ship (Wadam results).

The plots above show the three horizontal drift forces (forces along the X and Y axes, moment around the Z axis).

All three horizontal drift-load components show significant variations around $\omega = 0.7$ rad/s: a maximum for the X-force, a local maximum for the Z-moment, and a strong variation for the Y-force. It is interesting to note that the resonances for the RAOs of roll, heave, and pitch motions occurred at the same frequency. Drift forces are second-order forces that depend on the motions of the structure and the non-linear interactions between the waves and the surface. Therefore, when the RAOs reach a resonance, the amplified oscillations of the structure significantly alter the second-order components of the hydrodynamic forces, leading to substantial variations in the drift forces.

These mean drift loads primarily describe a demand on station keeping or heading control, rather than the vertical bending strength of the hull. If the vessel is restrained, the balancing forces must be transmitted through the relevant mooring or propulsion attachments into the surrounding structure. A freely drifting vessel would instead respond through its motion and hydrodynamic resistance. The results therefore identify an additional loading mechanism, but do not directly determine deck or bottom scantlings: that requires the spatial distribution of loads and the vessel's actual restraint conditions.

3.6 Section Analysis in Wadam

3.6.1 still water loads

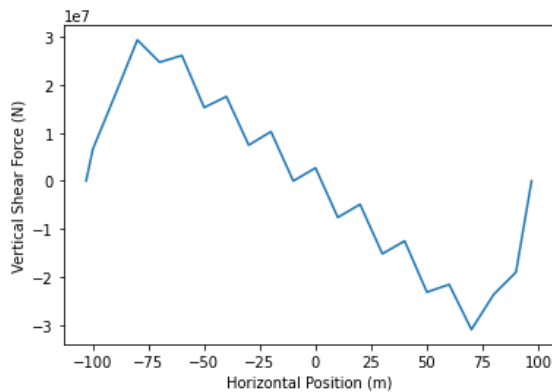


Figure 17: Vertical Shear Force Q_z along the ship

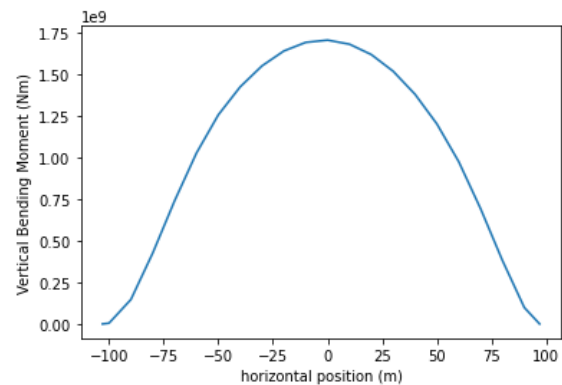


Figure 18: Vertical Bending Moment M_y along the ship

The reported maximum vertical shear force is equal to $Q_{zmax} = 29.38MN$ and the maximum vertical bending moment is equal to $M_{ymax} = 1707MNm$ (Sagging), in calm water. Total loads must combine still-water and wave-induced contributions at the same section with consistent signs; maxima from different sections cannot simply be added.

The reported still-water bending moment shows that the hull already carries substantial longitudinal loading before wave effects are added. Under sagging, the upper hull-girder structure is generally in compression and the bottom in tension, making the continuity of the strength deck, bottom structure and longitudinal stiffeners important. The compressed deck panels require attention to buckling as well as material strength. The vertical shear

force is carried mainly by the side shell and any effective longitudinal bulkheads, whose plating and stiffening must provide adequate shear resistance. These results also highlight the importance of cargo and ballast distribution, since changing the balance between weight and buoyancy changes the baseline structural demand.

3.6.2 Sectional loads in waves

The sectional loads F_z and M_y are calculated for a heading of 180° :

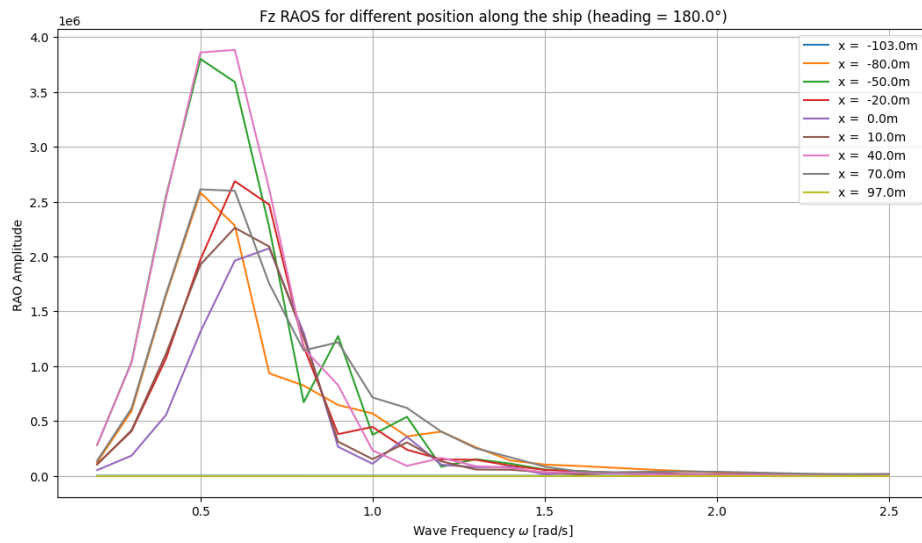


Figure 19: Amplitude of F_z for different x -position along the ship (heading = 180°)

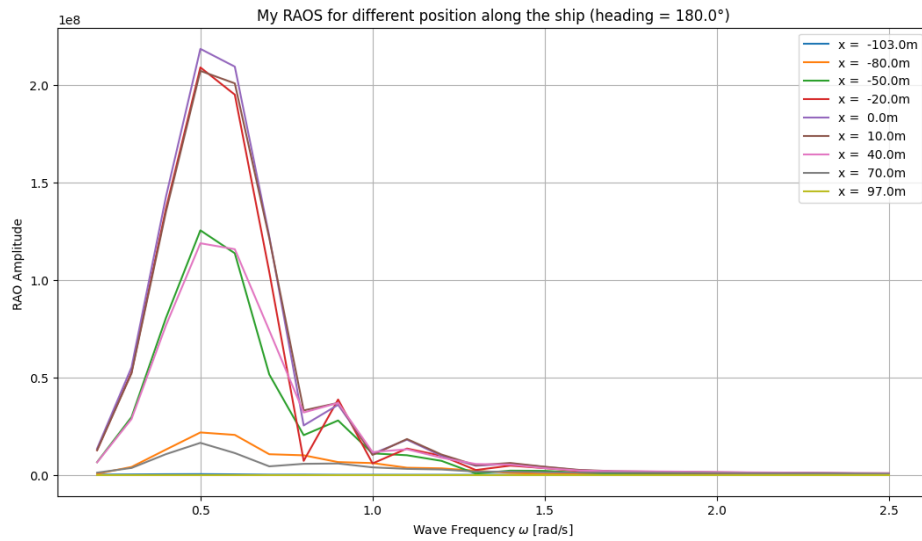


Figure 20: Amplitude of M_y for different x -position along the ship (heading = 180°)

These plots show that the maximum amplitudes occur at different locations along the ship.

For F_z , the maximum amplitude is reached at $x = -50$ m and $x = 40$ m.

For M_y , the maximum amplitude is reached around $x = 0$ m.

The maximum amplitudes for F_z and M_y are reached for $\omega = 0.5 \text{ rad/s}$

The different locations of the bending and shear maxima indicate that a single midship check would not capture all potentially critical sections. The bending peak near midship places particular emphasis on the longitudinal strength of the deck and bottom, whereas the shear peaks away from midship require attention to the side shell, longitudinal bulkheads and continuity of load transfer in those regions. Wave-induced bending also varies in sign, so both sagging and hogging combinations should be considered when superimposing wave loads on the still-water condition. The peak near 0.5 rad/s identifies a sensitive part of the regular-wave response; its importance in an irregular sea depends on the wave energy present at that frequency.

3.7 short and Long Term Analysis

Short- and long-term analyses are performed for the vertical shear force F_z and vertical bending moment M_y .

3.7.1 short term analysis

short term analysis for heading 180° for irregular waves defined by the PM spectrum with $H_s = 10 \text{ m}$ and $T_p = 15 \text{ s}$, during 3 hours on the cross section at midship ($x = 0 \text{ m}$).

	MPV	T_z
vertical shear force	$7.04E + 06$ N	11.20s
Vertical Bending Moment	$1.083E + 09$ Nm	13.43s

Table 1: short term results

For the selected three-hour sea state, the reported wave-induced bending maximum is of the same order as the reported still-water bending maximum, indicating that wave loading is a significant contribution to the hull-girder assessment. Their direct sum is not justified without confirming that both values refer to the same section and compatible signs. Structurally, the combined condition calls for checks of longitudinal yielding and compression-panel buckling, together with shear resistance. The response periods indicate repeated loading over the sea-state duration, but fatigue damage cannot be inferred from the maximum alone: stress ranges and cycle counts are required. These interpretations remain conditional on verification of the original extreme-value calculations.

3.7.2 long term analysis

The scatter diagram used represents the North Atlantic sea. The analysis is during 20 years for the cross section at midship.

	MPV	T_z
vertical shear force	1.702×10^7 N	9.00s
Vertical Bending Moment	1.953×10^9 Nm	9.94s

Table 2: long term results

The results obtained for the ship's cross sections could be used in a complementary study on the design of the cross section, by verifying that the stresses experienced in the cross sections due to these loads (shear force and bending moment) do not exceed the yield stress of the materials used in the ship.

The larger reported long-term extremes indicate that a structure assessed only for the selected three-hour sea state would not represent the full range of lifetime loading. For the hull girder, this motivates checking rare load combinations against yielding, buckling and ultimate strength, while assessing fatigue separately from the accumulated stress cycles over the operating profile. Global wave-load analysis supports both ultimate-strength and fatigue assessments [3]. The 20-year maximum is therefore an input to a structural assessment, not a direct measure of fatigue life or a prescribed plate thickness. The actual structural requirement depends on section geometry, material properties and load combinations, and the reported lifetime estimates remain subject to verification of the exposure model.

4 Ship in motion

Following the zero-speed assessment in calm water and irregular waves, the analysis is extended to forward speed using Wasim in HydroD.

The assessment examines the ship's RAOs and sectional forces and moments at speeds of 6, 8 and 10 m/s.

4.1 parameters in Wasim

4.1.1 simulation parameters

- The simulation duration is 300 seconds with a time step of 0.1 seconds. The ship accelerates to the desired speed using a ramp over 50 seconds.
- The water depth is 300 m.
- Motion control springs are defined to prevent the hull from leaving the testing area. The **damping coefficients** are set to 0.05 for all motions and the **Eigen periods** in the motion control springs parameters are defined as follows:
 - Surge = 100 s
 - Sway = 70 s
 - Yaw = 70 s
- Three directions are evaluated: 0°, 90°, and 180°.
- 16 frequencies are evaluated, ranging from 0.2 to 2.45 rad/s, with a step size of 0.15.
- The mass is defined using the mass distribution as explained previously 1(no point mass is applied).

These settings define the resolution and scope of the forward-speed assessment. The acceleration ramp reduces the abruptness of the initial transient, but a sufficiently settled response must still be identified before extracting amplitudes and phases. The relatively coarse frequency spacing may miss narrow response peaks, while the artificial motion-control springs may influence low-frequency motions. Consequently, the results support comparisons between the selected cases, but a structural design envelope would require sensitivity checks and additional headings.

4.1.2 Mesh

To run simulations on Wasim, it is necessary to define three meshes:

- The ship mesh,
- The free surface mesh,

- The control surface mesh to calculate the added wave resistance.

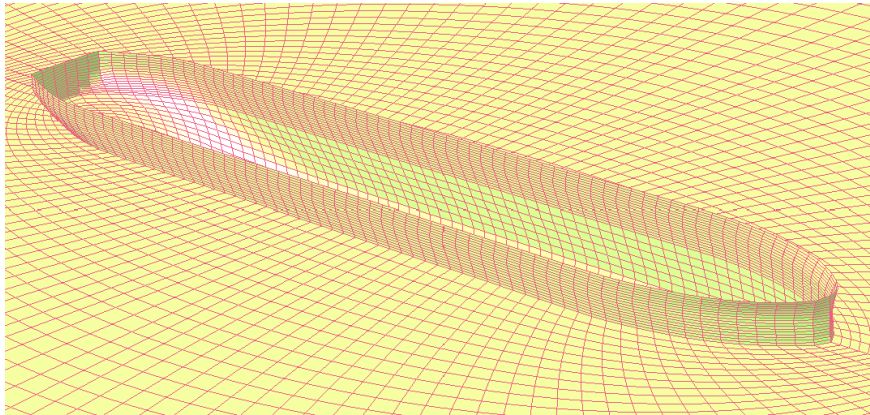


Figure 21: mesh of the hull and the free surface 1

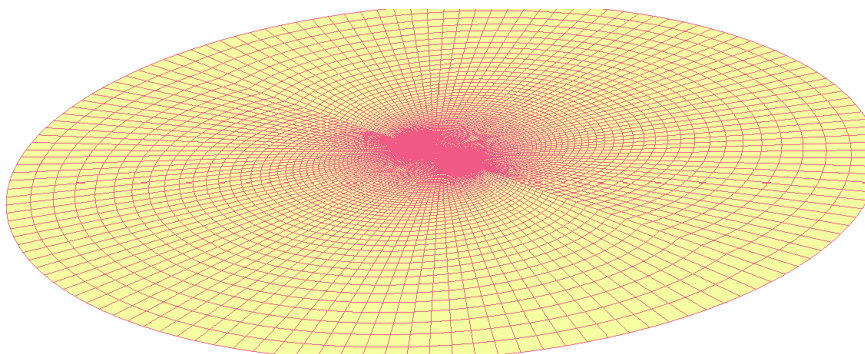


Figure 22: mesh of the hull and the free surface 2

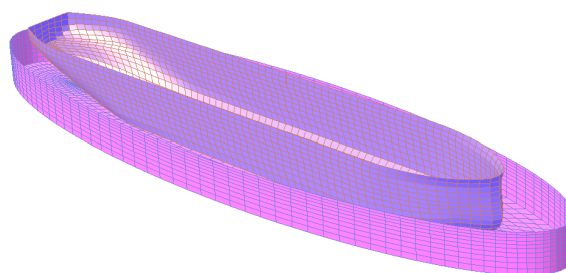


Figure 23: mesh of the hull and the control surface

The additional free-surface and control-surface meshes make the forward-speed results sensitive to more than the hull discretization alone. A credible load assessment requires the predicted motions and mean resistance to remain sufficiently stable when these meshes are refined or their extents are varied. From a structural perspective, this matters because

numerical changes in the hydrodynamic loading propagate into the forces and moments used for design. The mesh illustrations document the model arrangement, but do not by themselves establish convergence.

4.2 Motion response of the ship

4.2.1 RAOs

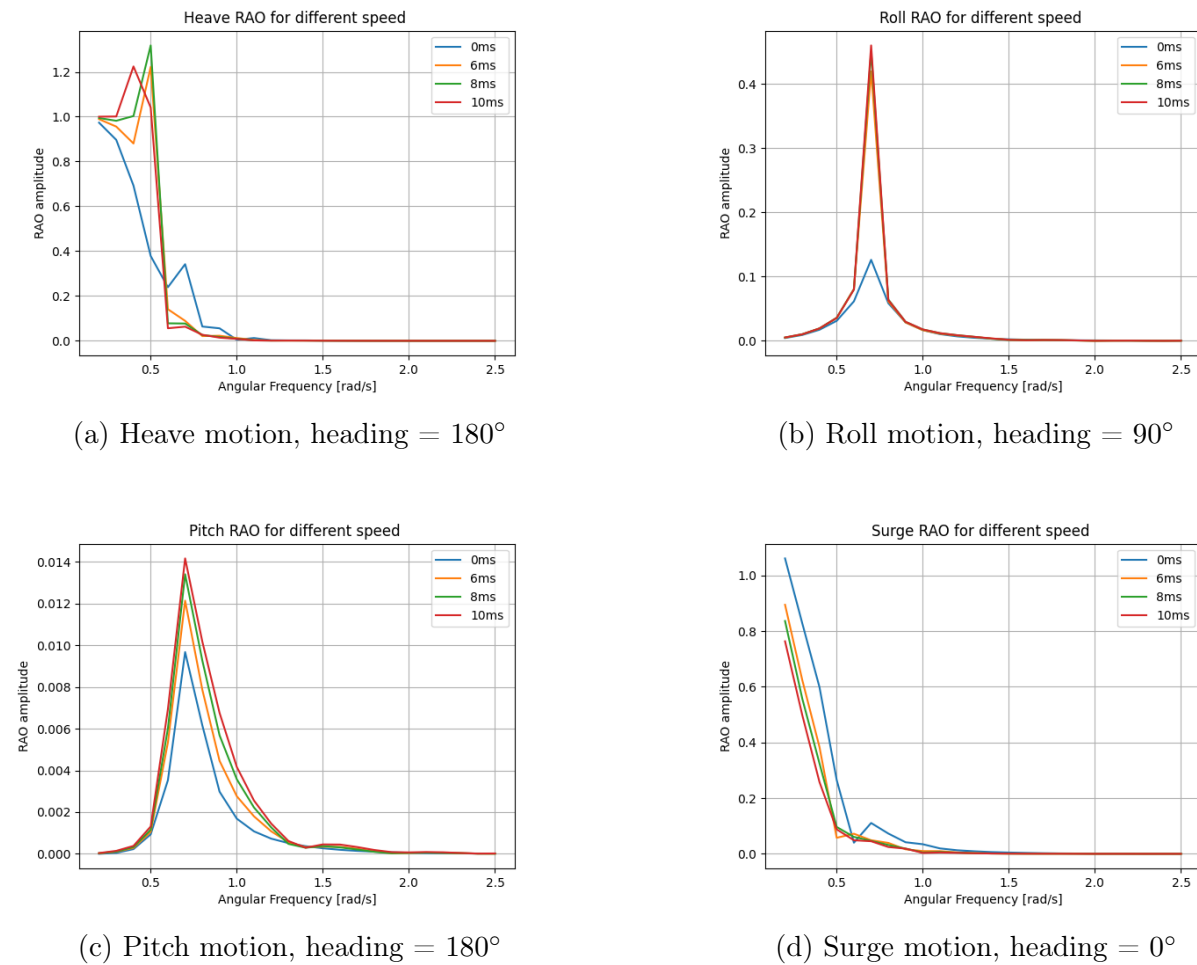


Figure 24: Different RAOS for the ship moving at 0, 6, 8 and 10 m/s.

The figures above provide information on the different motions of the ship for various headings, at speeds of 0 m/s (Wadam) and 6, 8, 10 m/s (Wasim). The following discussion examines each motion as a function of speed.

For propagation angle β and forward speed U , the signed encounter frequency is $\omega_e = \omega - kU \cos \beta$. Thus head seas increase the encounter frequency, while following seas can reduce

it; beam seas have no encounter-frequency shift. This relation alone does not determine RAO amplitudes.

- **Pitch:** the plotted peak amplitude increases with speed. The original discussion identifies head seas (180°), but its caption identified 90° . The exported plot does not state a heading, so the original simulation setting must be checked before assigning a definitive heading. Forward speed changes excitation and hydrodynamic coefficients as well as encounter frequency.
- **Roll:** the beam-sea curves at forward speed are similar. This is an observation for the selected cases, not a general independence of roll response from speed. Differences in the damping settings between Wadam and Wasim also affect the comparison. Oblique seas can excite roll.
- **Surge:** the following-sea plot shows a decrease in response with speed for the selected cases. A general monotonic trend cannot be inferred from these simulations alone.

The increase in the plotted pitch peak suggests a greater contribution of rotation to vertical motion near the vessel ends for the corresponding wave components. This can affect equipment accelerations and local wave-relative motions even when midship heave changes less markedly. However, a larger motion RAO does not necessarily imply a larger sectional bending moment: sectional loads depend on the spatial distribution and phase of both hydrodynamic and inertial forces. Motion and load results should therefore be interpreted together when identifying structurally demanding operating conditions.

4.2.2 Temporal Response

At forward speed, time reconstruction must use encounter frequencies and consistently phased complex RAOs. If the wave spectrum is transformed to encounter-frequency coordinates, conservation of spectral variance requires the appropriate Jacobian, including multiple branches where applicable in following seas. The retained time-history plot has not been regenerated from the corrected formulation.

The heave time history is reconstructed from a wave spectrum using the same approach as for the zero-speed Wadam analysis, with the forward-speed considerations stated above. The wave spectrum is a Pierson-Moskowitz spectrum with $H_s = 10$ m and $T_p = 15$ s.

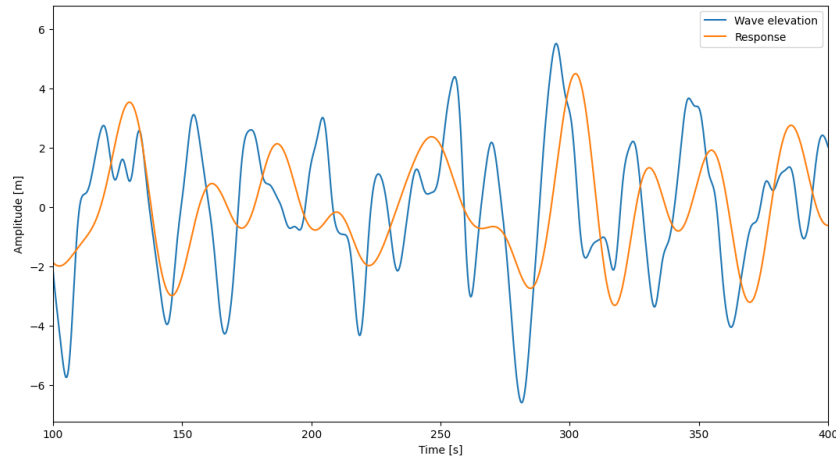


Figure 25: Heave temporal response for a ship moving at 10 m/s (19.4 knots)

The response above (Figure 25) clearly shows that the ship is sensitive to long waves (low frequency, long period), while small waves (high frequency, short period) have very little amplitude.

This time history illustrates the irregular sequence of motion cycles within one sea-state realization, rather than a single design amplitude. For structural interpretation, the associated accelerations are relevant to equipment supports and cargo securing, whereas hull-girder demand requires the corresponding sectional-load histories. Neither acceleration levels nor fatigue damage can be read directly from the heave displacement plot. A comparison with the zero-speed record would be more informative using consistent durations and statistical measures, since individual peaks also depend on the selected random wave phases.

4.2.3 Green water on the deck

Green water involves water passing over the deck. A preliminary linear screening compares local wave elevation with the moving deck at the same location. For small rotations, right-handed axes with x forward and z upward give

$$z_P(t) = \xi_3(t) + y_P \xi_4(t) - x_P \xi_5(t), \quad r_P(t) = \eta_P(t) - z_P(t). \quad (4.1)$$

The coordinates are relative to the motion origin, and pitch and roll are in radians. At a centreline stern point, $x_P < 0$; $x_P = -L/2$ is valid only when the origin is at midship. All motion and wave components must retain consistent phases, including the spatial phase of the wave at the evaluation point. Deck exceedance is indicated when r_P exceeds the local still-water freeboard, measured from the same vertical datum.

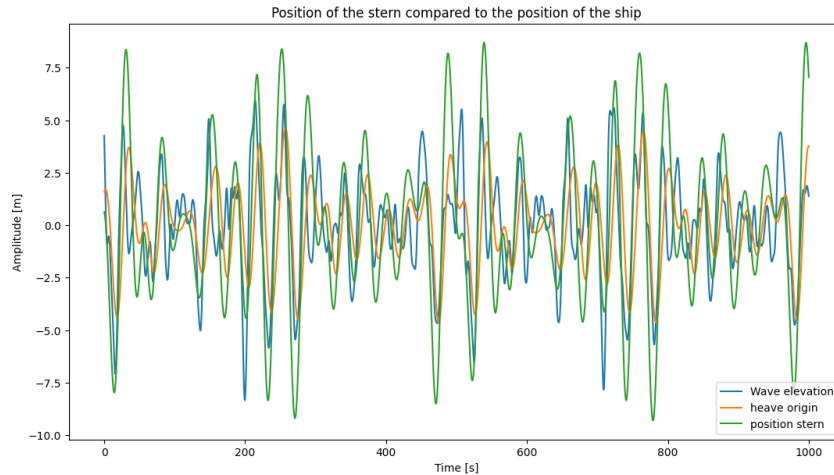


Figure 26: Original 1000 s time history labelled as a stern response at 10 m/s. The point coordinates and pitch-sign convention require verification.

The original calculation reports a maximum relative elevation of 6.18 m and a deck level of approximately 8 m. If 8 m is the local freeboard and the reported relative elevation uses the correct point, phase and sign conventions, this realization would not exceed the deck level. The available source does not establish those conditions: the original equation, prose and plot use inconsistent bow/stern descriptions. The result therefore remains a preliminary stern screening, not a verified bow assessment or a general conclusion that green water cannot occur. Local wave disturbance and nonlinear water-on-deck effects are outside this linear screening.

For structural design, deck wetness is primarily relevant to local loading on exposed deck plating, stiffeners, fittings and equipment foundations. An exceedance of freeboard would identify a condition requiring a separate assessment of water-on-deck pressures; it would not quantify those pressures. Conversely, the absence of an exceedance in a single linear realization does not demonstrate adequate resistance to rare local loads. The present screening therefore helps identify which locations and operating conditions merit further investigation, rather than specifying local reinforcement.

4.3 Short Analysis for the sectional loads

Only short-term results are presented at forward speed. The selected simulations do not provide the heading coverage and exposure model needed for the intended long-term assessment.

4.3.1 short term analysis

short term analysis for heading 180° for irregular waves defined by the PM spectrum with $H_s = 10m$ and $T_p = 15s$, during 3 hours for the cross section at midship ($x = 0m$).

speeds (m/s)	6	8	10
Vertical Shear Force	$9.163E + 06$ N	$9.628E + 06$ N	$1.045E + 07$ N
Vertical Bending Moment	$1.038E + 09$ Nm	$9.666E + 08$ Nm	$9.648E + 08$ Nm

Table 3: short term results for different ship speed

Again, this results could be used in a complementary study.

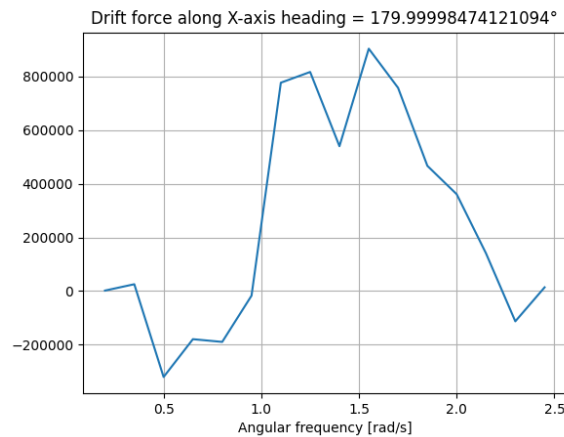
Within the forward-speed cases, the reported midship shear maximum increases from 6 to 10 m/s, while the bending maximum decreases slightly. This indicates that the most demanding speed depends on the load component: a condition that reduces bending demand can still increase the demand on shear-carrying side shell and longitudinal bulkheads. The deck and bottom should therefore be assessed for their governing bending combinations, and the shear-carrying structure for its own governing combinations. Since these results refer only to midship, they do not establish the maximum shear demand along the entire vessel.

4.4 Added resistance

The control-surface calculation estimates added wave resistance for the selected operating conditions. Its associated power demand is one contribution to propulsion requirements; calm-water resistance and other loads are not included here.

4.4.1 Quadratic Transfer Function

HydroD provides the ship's quadratic transfer function (QTF) for added resistance:

Figure 27: Drift force along the x-axis for a heading = 180° at 10m/s

Combining the QTF with the wave spectrum gives the mean drift force in the selected sea state.

The QTF identifies how individual wave-frequency components contribute to mean longitudinal loading. Its structural and operational significance depends on the overlap with the

sea-state spectrum: a pronounced QTF peak contributes little if the corresponding waves contain little energy. This weighting explains why the maximum ordinate of the QTF is not, by itself, an appropriate measure of propulsion demand or structural severity. The integrated mean force also does not describe fluctuating local pressures or the sectional bending response.

4.4.2 Averaged mean drift force

The previously defined wave spectrum is retained. The averaged mean-drift force in a seastate is defined as

$$\overline{F}_{avg} = 2 \int_0^\infty Q(\omega)S(\omega)d\omega \quad (4.2)$$

Here $Q(\omega)$ denotes the diagonal mean-drift QTF normalized so that a regular wave of amplitude A produces mean force $Q(\omega)A^2$, and S is a one-sided wave spectrum. The factor 2 follows from this normalization. The exported HydroD QTF normalization and force sign must be confirmed before using the numerical integral as a positive resistance.

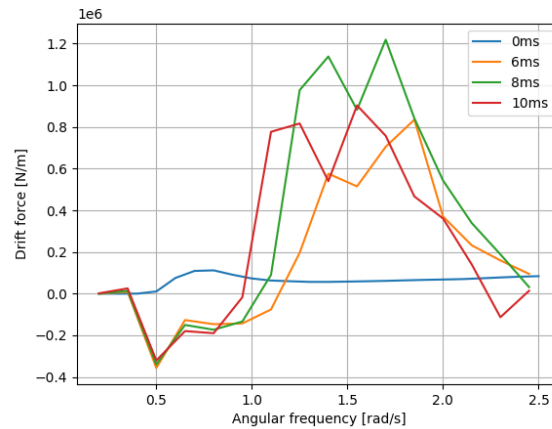


Figure 28: QTF in the x direction at 0, 6, 8 and 10 m/s

	0 m/s	6 m/s	8 m/s	10 m/s
$\overline{F}_{avg}$	0.2MN	1.28MN	1.33MN	1.32 MN

Table 4: Averaged mean-drift force for different ship speed

The similar mean forces at 6, 8 and 10 m/s do not by themselves demonstrate a convergence problem. Mesh refinement, time-step and duration sensitivity, and frequency-resolution checks are needed to assess convergence.

Using the reported added resistance at 10 m/s, $R_{AW} = 1.32$ MN, the associated effective power is

$$P_{E,AW} = R_{AW}U = 13.2 \text{ MW}. \quad (4.3)$$

With the assumed propeller efficiency 0.70 and mechanical efficiency 0.98, the illustrative overall efficiency is $\eta = 0.686$, giving

$$P_{B,AW} = \frac{13.2}{0.686} \approx 19.2 \text{ MW}. \quad (4.4)$$

This is an estimate of the additional engine power associated with the reported wave resistance, under the stated efficiency assumptions. It is not total installed engine power. Engine sizing also requires calm-water resistance, a consistent propulsion-efficiency model, other environmental loads and operating margins. The result does not establish that the vessel can safely escape a storm.

The nearly constant reported resistance across the three forward speeds does not imply a constant additional power requirement, because effective power also scales with speed. Maintaining the higher speed therefore places a greater demand on propulsion even when the mean opposing force changes little. The balancing thrust must be transmitted through the propulsion arrangement into the hull, making its supporting structure relevant to a subsequent design study. However, the mean resistance alone does not provide the local loads needed to size foundations or replace a hull-girder strength assessment.

4.5 Comparison with the zero-speed results

Here, zero speed refers to the Wadam analysis in waves, not to calm water. The most direct comparison uses the reported three-hour wave-induced maxima at midship for head seas, with $H_s = 10$ m and $T_p = 15$ s. The still-water maxima and the 20-year extremes represent different quantities and are excluded from this comparison. Table 5 converts the existing results to common units; percentage changes are relative to the zero-speed values and do not represent new simulations.

U (m/s)	Shear (MN)	Change	Bending (MN m)	Change
0	7.04	—	1083	—
6	9.163	+30.2%	1038	-4.2%
8	9.628	+36.8%	966.6	-10.7%
10	10.45	+48.4%	964.8	-10.9%

Table 5: Reported short-term midship wave-load maxima at zero and forward speed.

Sectional loads. Relative to the Wadam baseline, the Wasim results show approximately 30–48% higher shear maxima and 4–11% lower bending maxima. Structurally, the shear trend highlights the need to assess the side shell and effective longitudinal bulkheads at forward speed, even though the reported midship bending demand is lower. The latter does not imply an equivalent reduction in deck or bottom scantlings: still-water loading, other sections, headings and strength criteria may govern. A design comparison should therefore consider separate shear and bending envelopes rather than rank each speed using one response alone.

Motions and sectional demand. The increase in the plotted pitch peak alongside the lower short-term bending maximum is not necessarily contradictory. A motion peak at one frequency and a spectrum-weighted extreme sectional load describe different responses. Their relation depends on phase, frequency weighting and the distribution of hydrodynamic and inertial forces. The similar forward-speed roll curves also do not demonstrate unchanged structural loading, particularly because the zero-speed roll calculation uses different damping settings.

Mean longitudinal force. The reported mean force rises from 0.20 MN at zero speed to 1.28–1.33 MN at forward speed, approximately 6.4–6.7 times the baseline. This comparison concerns the reported mean drift-force magnitudes under the stated normalization. At zero speed, the force does not correspond to forward propulsive power, since $U = 0$; at forward speed, an opposing mean force contributes to added resistance and propulsion demand. The increase should therefore not be interpreted as the same multiplication of total engine power or hull-girder stress.

Scope of the comparison. These differences describe the available calculations, rather than isolating speed as their sole cause. The baseline uses Wadam and the moving cases use Wasim, with different numerical formulations, meshes, frequency grids and damping settings. A Wasim run at zero speed with matched inputs would help separate solver and modelling effects from forward-speed effects. The comparisons also remain conditional on checking the original extreme-value processing, frequency conventions, pitch heading and QTF normalization. Within that scope, the principal engineering finding is that forward speed redistributes demand between response components rather than increasing every load uniformly.

4.6 Status of the reported numerical results

The simulation plots and load tables are retained from the original project. Correcting the written equations does not establish which formulas were used in the original post-processing. Short- and long-term extreme loads require verification against the original scripts or solver outputs; the pitch heading, stern-point conventions and QTF normalization also remain to be checked. No new hydrodynamic simulations were performed for this revision.

References

- [1] MIT OpenCourseWare. *18.357 Interfacial Phenomena, Lecture 19: Water waves*. 2010. https://ocw.mit.edu/courses/18-357-interfacial-phenomena-fall-2010/229785489c81648500f89044a97c0e51_MIT18_357F10_Lecture19.pdf.
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